

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA10 / CSA1003	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	Total Marks:	25
CO1 Statement:	<i>Apply advanced methodologies and agile project management techniques to manage software projects effectively</i>		
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Date:	22/07/2026	Duration:	60 minutes

Code of Conduct

I am J. LOKESH-192311216 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: J. LOKESH

Smart Tax Calculation and Financial Planning Management System

1. Understand the Problem Statement

- Explain the problem in your own words.

The **Smart Tax Calculation and Financial Planning Management System** is developed to simplify tax calculation and improve financial management for both individuals and businesses. In today's world, calculating taxes manually is a difficult and time-consuming process because tax laws are complex and change frequently. Many people make mistakes while calculating taxes, which may result in incorrect tax payments, financial losses, or legal penalties. Traditional methods such as maintaining paper records or using spreadsheets increase the chances of human error and make financial management inefficient. In addition, users often struggle to organise their income, expenses, savings, and investments effectively, making it difficult to plan their finances.

Major Problems Identified

- Manual tax calculations are time-consuming and prone to errors.
 - Frequent changes in tax laws make calculations difficult.
 - Paper records and spreadsheets are difficult to manage.
 - Users cannot easily track income, expenses, savings, and investments.
 - Lack of financial planning leads to poor budgeting and investment decisions.
 - Professional financial advisors may not always be available or affordable.
 - Sensitive financial information requires strong security and privacy protection.
- Identify the objectives of the proposed system.

The **Smart Tax Calculation and Financial Planning Management System** is designed to simplify tax calculation and improve financial management for individuals and businesses. The primary objective of the system is to automate tax calculations using the latest tax regulations, thereby reducing manual effort and improving accuracy. It also helps users manage their financial records efficiently, plan their budgets, and make informed financial

decisions. The system provides a secure and user-friendly platform that supports effective tax management and long-term financial planning.

Objectives of the Proposed System

- Automate tax calculation based on the latest government tax rules.
 - Reduce manual errors and improve the accuracy of tax calculations.
 - Maintain organised records of income, expenses, savings, and investments.
 - Help users create budgets and achieve their financial goals.
 - Provide personalised tax-saving and investment recommendations.
 - Generate accurate tax reports and financial statements.
 - Ensure secure storage and protection of users' financial data.
 - Send reminders for tax filing deadlines and important financial activities.
 - Enable administrators to manage users and update tax regulations easily.
 - Improve overall financial planning, decision-making, and compliance with tax laws.
- State the scope of the problem.
- The **Smart Tax Calculation and Financial Planning Management System** is designed to provide a complete solution for tax calculation and financial management. The scope of the system includes managing financial records, automating tax calculations, generating reports, and supporting financial planning for both individuals and businesses. It focuses on improving accuracy, reducing manual effort, and ensuring compliance with the latest tax regulations. The system also provides a secure platform for storing financial information and assisting users in making better financial decisions.

2. Identify Key Stakeholders and Challenges

- Identify all the stakeholders involved in the system.

1. Individual Users (Taxpayers)

- Register and use the system for tax calculation.
- Manage personal income, expenses, savings, and investments.

- Generate tax reports and financial statements.

2.Business Owners

- Maintain business financial records.
- Calculate business taxes accurately.
- Monitor company income, expenses, and profits.

3.Financial Advisors

- Guide users in financial planning.
- Recommend investment and budgeting strategies.
- Help users achieve financial goals.

4.Tax Consultants

- Verify tax calculations.
- Assist users in tax filing.
- Ensure compliance with government tax regulations.

5.Government Tax Department

- Establish tax rules and policies.
- Update tax regulations.
- Monitor tax compliance and legal requirements.

6.System Administrator

- Manage user accounts and permissions.
- Update tax rules in the system.
- Monitor system performance and security.
- Perform database backup and recovery.

7.Software Developers

- Design and develop the application.

- Implement new features and enhancements.
- Fix software bugs and improve performance.

8. Testing Team

- Test all system modules.
- Identify and report software defects.
- Verify functionality, security, and performance.

9. Maintenance Team

- Perform regular software updates.
- Resolve technical issues.
- Maintain system reliability and security.
- Ensure continuous system availability.

- Explain the roles and responsibilities of each stakeholder.

1. Individual User

- Registers and logs into the system securely.
- Enters accurate income, expense, savings, and investment details.
- Uses the system to calculate taxes and generate financial reports.
- Reviews tax calculations and plans personal finances.

2. Business Owner

- Maintains business financial records.
- Calculates business taxes using the system.
- Monitors company income, expenses, and profits.
- Ensures compliance with tax regulations.

3. Financial Advisor

- Analyses users' financial information.

- Provides budgeting and investment advice.
- Recommends tax-saving strategies.
- Helps users achieve financial goals.

4.Tax Consultant

- Verifies tax calculations and reports.
- Assists users in tax filing.
- Ensures compliance with government tax laws.
- Advises users on tax-related matters.

5.Government Tax Department

- Defines tax rules and regulations.
- Updates tax policies periodically.
- Ensures legal compliance.
- Provides official tax guidelines followed by the system.

6.System Administrator

- Manages user accounts and permissions.
- Updates tax rules in the system.
- Monitors system performance.
- Maintains data security and performs regular backups.

7.Software Developers

- Design and develop the application.
- Implement new features and system modules.
- Fix software bugs and improve performance.
- Maintain and upgrade the system regularly.

8.Testing Team

- Tests all modules and system functionalities.
- Identifies and reports software defects.
- Ensures the system meets quality standards.
- Verifies security and performance before deployment.

9.Maintenance Team

- Performs regular software updates.
- Monitors system health and availability.
- Resolves technical issues.
- Ensures long-term reliability and security of the application.

- Discuss the major challenges and issues faced by the stakeholders.

1.Individual User

- Difficulty in understanding complex tax rules.
- Manual calculation errors.
- Lack of financial planning knowledge.
- Concerns about data privacy and security.

2.Business Owner

- Managing large volumes of financial data.
- Frequent changes in tax regulations.
- Risk of incorrect tax filing and penalties.
- Time-consuming financial record management.

3.Financial Advisor

- Analysing large amounts of client financial data.
- Keeping up with changing tax policies.
- Providing accurate and personalised financial advice.

- Managing multiple clients efficiently.

4.Tax Consultant

- Interpreting complex tax laws correctly.
- Ensuring accurate tax calculations.
- Handling tight tax filing deadlines.
- Avoiding legal and compliance issues.

5.Government Tax Department

- Updating tax policies regularly.
- Monitoring taxpayer compliance.
- Preventing tax fraud and evasion.
- Managing large volumes of taxpayer information.

6.System Administrator

- Ensuring system security and data protection.
- Managing user accounts and permissions.
- Maintaining system availability with minimal downtime.
- Performing regular backups and software updates.

7.Software Developers

- Developing accurate tax calculation algorithms.
- Integrating new features without affecting existing functions.
- Fixing software bugs and improving performance.
- Adapting the system to changing user requirements.

8.Testing Team

- Identifying all software defects before deployment.
- Testing system performance under heavy user load.
- Verifying security and data integrity.
- Ensuring all modules function correctly.

9.Maintenance Team

- Updating the software based on new tax regulations.
- Resolving technical issues quickly.
- Maintaining system performance and reliability.
- Ensuring continuous system availability and security.

3. Analyze the Current Approach

- Describe the existing system or current process.

Current Process

- Tax calculations are performed manually using calculators, spreadsheets, or paper records.
- Individuals and businesses maintain financial records separately.
- Income, expenses, savings, and investments are recorded manually.
- Tax reports are prepared using traditional accounting methods.
- Financial planning mainly depends on personal knowledge or professional advisors.

Features of the Existing System

- Manual entry of financial data.
- Paper-based or spreadsheet-based record management.
- Basic accounting software with limited automation.
- Manual calculation of tax deductions and exemptions.
- Limited tracking of financial activities.

Limitations of the Existing System

- Time-consuming and labour-intensive process.
- High possibility of human errors in tax calculations.
- Frequent tax law changes require manual updates.
- Financial records are often scattered and difficult to manage.
- No automatic tax-saving or investment recommendations.

- Identify its strengths and limitations.

The existing tax calculation and financial planning system has a few strengths but also several limitations. One of its main strengths is that it is simple to use for basic tax calculations and

financial record management, especially for individuals or small businesses with limited financial transactions. Manual verification by tax consultants or financial advisors can also improve the accuracy of tax calculations and ensure compliance with tax regulations. Additionally, traditional methods such as spreadsheets and accounting software require minimal technical knowledge and have a low initial implementation cost.

- Analyse the problems or gaps that need to be addressed.

The existing tax calculation and financial planning process has several problems and gaps that reduce its efficiency and reliability. One of the major issues is the dependence on manual calculations, which increases the chances of human errors and inaccurate tax computations. Frequent changes in tax laws and regulations make it difficult for individuals and businesses to stay updated, resulting in incorrect tax filing and possible penalties.

4. Propose a Suitable Solution Using Software Engineering Principles



- Design a software-based solution for the given problem.

A **Smart Tax Calculation and Financial Planning Management System** can be developed as a secure web-based and mobile-enabled application to automate tax calculation and financial management. The system allows users to register and log in securely, after which they can enter and manage their income, expenses, savings, investments, and other financial details. Based on the latest government tax rules and regulations, the system automatically calculates the user's tax liability, identifies eligible deductions, and generates accurate tax reports.

- Describe the major functional modules of the proposed system.

1. User Authentication Module

- Provides secure user registration and login.
- Manages user profiles and authentication.
- Ensures authorised access to the system.

2. Income and Expense Management Module

- Records income, expenses, savings, and investments.
- Allows users to update and manage financial records.
- Tracks financial transactions efficiently.

3. Tax Calculation Module

- Automatically calculates tax based on current tax regulations.
- Applies deductions and exemptions accurately.
- Generates tax summaries and tax reports.

4. Financial Planning Module

- Helps users create budgets and financial goals.
- Provides tax-saving suggestions.
- Recommends suitable investment options.

5. Report Generation Module

- Generates tax reports and financial statements.
- Provides income and expense summaries.
- Allows users to download and print reports.

6. Notification Module

- Sends reminders for tax filing deadlines.
- Notifies users about financial activities.
- Alerts users about updates in tax regulations.

7. Administrator Module

- Manages user accounts and permissions.
- Updates tax rules and system settings.
- Monitors system performance and security.

8. Security and Data Management Module

- Encrypts sensitive financial data.
- Performs database backup and recovery.
- Protects the system from unauthorised access and cyber threats.

- Identify the functional and non-functional requirements.

The **Smart Tax Calculation and Financial Planning Management System** has both **functional** and **non-functional requirements** to ensure that it operates effectively and meets user needs. The **functional requirements** define the core features of the system. These include user registration and secure login, profile management, recording and updating income and expense details, automatic tax calculation based on the latest tax regulations, budgeting and financial planning, investment and tax-saving recommendations, report generation, notification of tax filing deadlines, and administrator functions such as user management, tax rule updates, and system monitoring.

- Explain how software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security are incorporated into your solution.

The proposed **Smart Tax Calculation and Financial Planning Management System** is designed by applying important software engineering principles to ensure high quality and efficiency. **Modularity** is achieved by dividing the system into separate modules such as user management, tax calculation, financial planning, report generation, and administration. Each module performs a specific function independently, making the system easier to develop and maintain. **Scalability** is incorporated by designing the software to support an increasing number of users and future enhancements without affecting system performance

5. Justify the Chosen Methodology/Framework

- Select an appropriate Software Development Life Cycle (SDLC) model or software development framework (e.g., Agile, Scrum, Waterfall, Spiral, DevOps).

The **Agile Software Development Life Cycle (SDLC) model** is the most suitable approach for developing the **Smart Tax Calculation and Financial Planning Management System**. Agile focuses on developing the software in small and manageable iterations called sprints. Each sprint delivers a working version of the system with new features and improvements. This approach allows developers to respond quickly to changing tax laws and financial regulations. Agile encourages continuous communication between developers, stakeholders, and end users throughout the project. Regular feedback helps identify issues early and improves the quality of the software.

- Justify why the selected methodology/framework is suitable for the proposed system.

The **Agile SDLC model** is suitable for the **Smart Tax Calculation and Financial Planning Management System** because it supports continuous development and improvement. The system requires frequent updates to accommodate changing tax laws and government regulations. Agile allows these changes to be implemented quickly without affecting the entire application. The methodology divides the project into small iterations called sprints, making development more manageable. Each sprint delivers a working version of the software with new features and improvements. Regular feedback from users and stakeholders helps ensure that the system meets their requirements.

- Explain how it supports successful project development and maintenance.

The **Agile SDLC model** supports successful project development by dividing the project into small and manageable sprints. Each sprint focuses on developing and testing specific features of the system. This approach allows the development team to deliver working software at regular intervals. Continuous feedback from users and stakeholders helps improve the system after every sprint. Agile enables developers to identify and fix errors early, reducing the cost of maintenance. Regular testing throughout the development process improves software quality and reliability. The model also supports quick implementation of changes in tax laws and financial regulations.

6. Discuss Expected Outcomes and Limitations

- Describe the expected benefits of the proposed solution.

1. Improved Accuracy and Efficiency

- Automates tax calculations using the latest tax regulations.
- Reduces manual errors and saves time.
- Generates accurate tax reports quickly.

2. Better Financial Management

- Helps users manage income, expenses, savings, and investments.
- Supports budgeting and financial planning.
- Provides personalised tax-saving and investment suggestions.

3. Enhanced Security and Reliability

- Protects sensitive financial data through authentication and encryption.
- Ensures secure storage and data privacy.
- Maintains reliable system performance with regular backups.

4. User Convenience and Future Growth

- Offers a simple and user-friendly interface.
- Sends reminders for tax filing deadlines and financial activities.

- Supports future enhancements and adapts to changing tax regulations.
- Explain how the solution addresses the identified challenges.

The proposed **Smart Tax Calculation and Financial Planning Management System** effectively addresses the challenges identified in the existing system. It automates tax calculations, eliminating the need for manual computations and reducing human errors. The system uses updated tax rules to ensure accurate and compliant tax calculations. It provides a centralized platform for managing income, expenses, savings, and investments. This eliminates the problem of maintaining scattered financial records. The budgeting and financial planning features help users manage their finances more efficiently.

- Discuss potential risks, implementation challenges, and limitations.

The proposed **Smart Tax Calculation and Financial Planning Management System** may face several risks and implementation challenges. One major risk is the frequent change in government tax laws, which requires regular system updates. Cybersecurity threats such as hacking, phishing, and data breaches may compromise sensitive financial information. The system also depends on accurate user input, and incorrect data may produce inaccurate tax calculations. Integrating the software with external financial or banking systems can be technically challenging. The initial development and deployment costs may be high for the organisation. Users with limited technical knowledge may require training to use the system effectively.

- Suggest possible future enhancements.

The proposed **Smart Tax Calculation and Financial Planning Management System** can be enhanced with several advanced features in the future. Artificial Intelligence can be integrated to provide personalised financial planning and tax-saving suggestions. Machine Learning algorithms can analyse spending patterns and predict future financial trends. The system can be integrated with banks to automatically import transaction details. Mobile applications for Android and iOS can be developed to improve accessibility. Cloud-based storage can be implemented to provide secure access from anywhere. Multi-language support can be added to make the system accessible to users from different regions.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Understanding	Clearly explains the problem, objectives, and scope with complete understanding.	Explains the problem with minor omissions.	Basic understanding with limited explanation.	Poor understanding; objectives and scope are unclear or missing.
Stakeholder & Challenge Analysis	Identifies all stakeholders, their roles, and challenges with clear analysis.	Identifies most stakeholders and challenges with adequate analysis.	Identifies only a few stakeholders or challenges.	Stakeholders and system analysis are incomplete or inaccurate.
Current System Analysis	Thoroughly analyzes the existing system, strengths, weaknesses, and identifies all gaps.	Good analysis with most strengths and weaknesses identified.	Limited analysis with partial identification of issues.	Proposed solution is unclear or inappropriate; requirements are largely missing.
Proposed Software Solution & Software Engineering Principles	Proposes an innovative, feasible solution applying appropriate software engineering principles (modularity, scalability, maintainability, security, etc.).	Suitable solution with good application of software engineering principles.	Basic solution with limited application of software engineering principles.	Methodology is inappropriate, poorly justified, or not discussed.
Methodology/ Framework Justification	Selects the most suitable SDLC/model/framework and provides strong technical justification.	Suitable methodology selected with adequate justification.	Methodology selected but justification is weak.	Outcomes, limitations, or future enhancements are missing; report lacks organization and clarity.

